

h.1

$$z = ax + by + c$$

$$\begin{aligned} & (0, 0, 1) \\ & (1, 0, 1) \\ & (0, 1, 1) \\ & (1, 1, 2) \\ & (1, -1, 0) \\ & (-1, 1, 1) \end{aligned}$$

součet čtverců minimalní

→ metoda nejmenších čtverců

$$\left(\begin{array}{ccc|c} x_1 & y_1 & 1 & z_1 \\ \vdots & \vdots & \vdots & \vdots \\ x_n & y_n & 1 & z_n \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{array} \right)$$

A

$$\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{array} \rightarrow \begin{array}{ccc} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{array}$$

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

podle tvaru 8.91:

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$\xrightarrow{1} \begin{pmatrix} 4 & -1 & 2 & | & 2 \\ -1 & 4 & 2 & | & 4 \\ 2 & 2 & 6 & | & 6 \end{pmatrix} \xrightarrow{2} \begin{pmatrix} 0 & 15 & 10 & | & 18 \\ -1 & 4 & 2 & | & 4 \\ 2 & 2 & 6 & | & 6 \end{pmatrix}$$

$$\sim \begin{pmatrix} 0 & 15 & 10 & | & 18 \\ -1 & 4 & 2 & | & 4 \\ 0 & 10 & 10 & | & 14 \end{pmatrix} \sim \begin{pmatrix} 1 & -4 & -2 & | & -4 \\ 0 & 5 & 5 & | & 7 \\ 0 & 15 & 10 & | & 18 \end{pmatrix} \cdot (-3) \downarrow$$

$$\sim \begin{pmatrix} 1 & -4 & -2 & | & -4 \\ 0 & 5 & 5 & | & 7 \\ 0 & 0 & 5 & | & 3 \end{pmatrix} \quad \begin{pmatrix} 2/5 \\ 4/5 \\ 3/5 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2/5 \\ 4/5 \\ 3/5 \end{pmatrix}$$

$$\boxed{z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}}$$